

SAAIM ABDULLAH

Backend, Data & Cloud Software Engineer

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SUMMARY

Backend and data engineer (2+ years) building production APIs, event-driven ETL pipelines, and AWS-backed services. Delivered backend systems serving 1,000–3,000 daily users, real-time streaming pipelines (Kafka, Spark, Airflow), and RAG/ML services in Python and Django. Strong in REST API design, data warehousing, and containerized cloud deployment on AWS.

EXPERIENCE

[Fitter Health](#)

May 2026 - Jul 2026

Software Engineer (Backend / Data)

Spain / Remote

- Built backend systems and API services for a preventive health-tech platform using **Django** and **Next.js**, focused on scalable, maintainable product modules.
- Developed cloud-backed services on **AWS** using Redshift, SNS, SQS, ECS, RDS, S3, API Gateway, IAM, and VPC.
- Defined API contracts and backend workflows to improve product reliability, maintainability, and readiness for data-driven features.

[Expertflow – Cisco UCCE Solution Partner](#)

Jun 2024 – Feb 2026

Software Engineer (Backend / Cloud)

Lahore, Pakistan

- Built backend APIs serving **1,000–3,000 daily users** via scalable AWS-backed services, deployed as **Docker** containers on **AWS EC2** with IAM, VPC, and S3; configured delivery behind an **Application Load Balancer (ALB)** with rate-limiting and traffic controls.
- Designed and shipped event-driven batch and streaming ETL pipelines on AWS using **Lambda**, **MWAA (Airflow)**, **EMR**, with **PySpark**, Glue, and S3 automating ingestion, transformation, and cataloging to deliver analytics-ready Parquet datasets for BI reporting and ML feature workflows..
- Built transformation pipelines to standardize raw inputs, improving data quality and usability across systems.
- Developed a real-time voice/video **communication SDK (WebRTC)** and RESTful APIs integrating frontend clients, SDK modules, and third-party platforms.
- Implemented a chatbot workflow to improve first-response time, and delivered SQL-based analytics for actionable customer insights.
- Enhanced platform security via time-bound access controls, and collaborated across **Angular** frontend, backend, and SDK teams to deliver end-to-end product features.

PROJECTS

[Real-Time E-Commerce ETL Pipeline](#) | *Kafka, PySpark, Airflow, Postgres* [Article](#)

2026

- Built a **medallion-architecture** (bronze/silver/gold) streaming pipeline moving Kafka-published orders into a Parquet bronze layer on S3-compatible storage via PySpark Structured Streaming.
- Modeled a gold **star schema** (customer, product, date dimensions around a sales fact table) and orchestrated batch stages with an **Airflow DAG** on Docker Compose.

[Multilingual, Multi-Tenant RAG Engine](#) | *Django, PostgreSQL, Local Embeddings, RAG* [Article](#)

2026

- Built a production-style **RAG** backend with tenant-isolated retrieval, local multilingual embeddings, and grounded generation for a localization Q&A service on Django + PostgreSQL 18.
- Documented setup, database friction, and fixes in a companion article on running a Django RAG backend on PostgreSQL 18.

[Movie Recommender System](#) | Python, FastAPI, scikit-learn, Airflow, Redis, Docker

Ongoing

- Building a production-ready movie recommendation service using MovieLens interaction data, item metadata, and modular training/serving pipelines.
- Implemented a hybrid approach combining content-based similarity, item co-occurrence signals, and popularity bias for cold-start stability.
- Developed **FastAPI** endpoints for health checks, personalized recommendations, similar-item retrieval, and model reloading; separated ingestion, feature engineering, training, evaluation, and serving layers.
- Added Docker Compose, optional Redis caching, API-key protected endpoints, structured logging, request tracing, and CI-ready testing workflows.

[Autoencoder Compression for Edge Devices](#) | Python, TensorFlow, TF Lite Micro, Flask, AWS

2025

- Developed an autoencoder to compress and reconstruct sensor data for lightweight edge-device communication, tuning latent-vector size for compact representation.
- Converted the encoder for microcontroller-side inference and exposed the decoder through a **Flask API** on AWS EC2 with S3 integration for encoded payload handling.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java
Backend & APIs: Django, Django REST Framework, FastAPI, Flask, Node.js, Express.js, REST APIs, WebRTC, API Design
Data Engineering: ETL Pipelines, Apache Kafka, PySpark (Structured Streaming), Apache Airflow, Pandas, NumPy, Data Warehousing, Star Schema, Medallion Architecture
Cloud / DevOps: AWS (Lambda, EC2, ECS, S3, RDS, Redshift, SQS, SNS, API Gateway, IAM, VPC, ALB), Docker, Docker Compose, Git, CI/CD
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Firebase
ML / AI: scikit-learn, TensorFlow, TensorFlow Lite Micro, RAG (Retrieval-Augmented Generation), Recommender Systems, Multilingual Embeddings
Frontend: React, Next.js, Angular, HTML/CSS

EDUCATION

COMSATS University Islamabad, Lahore Campus <i>Bachelor of Science in Computer Engineering</i> Relevant coursework: Data Structures, Database Management, Computer Networks, Operating Systems, Software Engineering, Computer Architecture, Machine Learning.	2021 – 2025 <i>Lahore, Pakistan</i>
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CERTIFICATIONS

AWS Cloud Technical Essentials <i>Amazon Web Services, Coursera</i>	Mar 2026
Introduction to Data Engineering <i>DeepLearning.AI, Coursera</i>	Feb 2026
Getting Started with Git and GitHub <i>IBM, Coursera</i>	Feb 2026